

R RACKYMUTHU

Manager, Deloitte | Python · Generative AI · Cloud · DevOps | 10.2 Years of Experience

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PROFESSIONAL SUMMARY

Engineering Manager at Deloitte with 10.2 years of experience building scalable microservices, cloud-native platforms, and AI-driven enterprise applications for global clients including Experian (US) and Sky (UK). Deep expertise across AWS, Azure, and GCP with strong DevOps practice — Terraform IaC, Kubernetes, CI/CD, and full-stack observability. Growing focus on Generative AI, including LLM integration, RAG pipelines, and document intelligence. Proven record of leading teams, engaging clients directly, and owning solutions from requirements through production.

KEY HIGHLIGHTS

- Architecting a real-time analytical feature-store platform (Feature Builder) for Experian using FastAPI, Kafka, and Kubernetes on AWS — a centralized hub used by data scientists and engineers across projects.
- Generative AI practitioner: building LLM-powered solutions including document intelligence, RAG pipelines, and prompt-engineered workflows — backed by early ML experience delivering an AI chatbot in production.
- Multi-cloud specialist: delivery across AWS, Azure, and GCP with infrastructure fully automated via Terraform.
- Advanced DevOps practitioner: blue-green and canary deployments, ArgoCD GitOps, and end-to-end observability with Dynatrace and OpenTelemetry.
- Direct client ownership with US and UK enterprises — requirement gathering, agile delivery, and stakeholder communication.
- Team leader and mentor: trained freshers, guided junior engineers, and led an end-to-end ERP implementation as project lead.
- Full-ownership delivery: built complete products solo — SSO identity platform, hostel management system — from conceptualization to deployment.

TECHNICAL SKILLS

Languages	Python, JavaScript, SQL, Groovy
Backend	Django, Flask, FastAPI, REST & GraphQL APIs, Microservices Architecture
Generative AI	LLM Integration, Agentic AI & AI Agents, RAG Pipelines, LangChain/LangGraph, MCP, AWS Bedrock & Azure OpenAI, Prompt Engineering, Function/Tool Calling, Embeddings & Semantic Search, LLMOps, Fine-tuning (LoRA/PEFT), Vector Databases
Frontend	React
Databases	MySQL, PostgreSQL, MongoDB, Redis, DynamoDB, BigQuery
Cloud	AWS (EC2, S3, Lambda, RDS, EKS, ECS, SageMaker, IoT Core, CloudFormation), Azure (Functions, IoT Hub, ML, App Services), GCP (Vertex AI, BigQuery, Pub/Sub)
DevOps / CI-CD	Docker, Kubernetes, Helm, Jenkins, GitLab CI/CD, ArgoCD, Terraform, Ansible, OpenShift, Blue-Green & Canary Deployments
Observability	Dynatrace, OpenTelemetry, Prometheus, Grafana, Kibana, Zabbix, Splunk
Other	Kafka, RabbitMQ, Keycloak, OpenLDAP, ERPNext/Frappe, Metabase, Agile/Scrum, Confluence

WORK EXPERIENCE

Manager — Deloitte India Coimbatore, Tamil Nadu	Sep 2022 – Present
Feature Builder — Analytical Platform · Client: Experian – Developed scalable microservices and RESTful APIs for creating, managing, and tracking ML features in real time on a centralized feature-store platform.	Jul 2024 – Present

- Ensured seamless cross-application integration and high-performance data processing in collaboration with cross-functional teams.
- Drove scalability, performance optimization, and testing to maintain platform stability; documented backend processes for knowledge sharing.
- Integrated Agentic AI capabilities — LLM-powered agents with tool calling that automate feature discovery, validation, and recommendation workflows across the platform.
- Built agent orchestration with LangGraph and connected agents to internal services and data sources via MCP (Model Context Protocol).
- Leveraged AWS Bedrock for scalable LLM inference, aligning Gen AI workloads with the platform's existing AWS-native architecture.

Tech Stack: Python, FastAPI, PostgreSQL, MongoDB, Kafka, Docker, Kubernetes, ArgoCD, Terraform, Dynatrace, AWS Bedrock, LangGraph, MCP, Agentic AI

► **Collective IQ — Project Management & Analytics Platform** · Client: Experian *Sep 2022 – Present*

- Implemented core application modules for a specification-driven project platform with syntax validation and multi-language program logic.
- Engaged directly with the US client to gather requirements; mentored and guided junior team members.
- Provisioned and automated AWS infrastructure with Terraform, streamlining deployment and scalability.
- Introduced LLM-assisted specification authoring and validation using LangChain with Azure OpenAI Service, reducing manual syntax-review effort.
- Prototyped AI agents with tool calling to automate project setup and validation distribution workflows.

Tech Stack: Flask, MySQL, React, Terraform, Tanium, Dynatrace, AWS (EC2, S3, RDS, CloudWatch), LangChain, Azure OpenAI, Agentic AI

Senior Software Engineer — Tata Consultancy Services

May 2021 – Sep 2022

Bangalore, Karnataka

► **Sauran — Real-Time API Management** · Client: Sky UK

May 2021 – Sep 2022

- Built an API management application handling real-time network traffic data, with Swagger-documented endpoints and traffic monitoring.
- Worked directly with UK stakeholders in an agile framework — owning development from inception to delivery with regular feedback cycles.
- Implemented embeddings-based semantic search over API documentation and traffic metadata, improving endpoint discoverability.
- Engineered LLM function/tool-calling integrations to automate API diagnostics and traffic-analysis queries.
- Applied LLMops and AI observability practices — monitoring model behavior and drift alongside application telemetry.
- Experimented with parameter-efficient fine-tuning (LoRA/PEFT) to adapt models to network-traffic domain terminology.

Tech Stack: Python, Flask, MySQL, BigQuery, Embeddings & Semantic Search, LLM Function Calling, LLMops, LoRA/PEFT

Software Engineer — Digixr Technologies

Feb 2020 – May 2021

Coimbatore, Tamil Nadu

► **SMOOR Chocolate ERP** · Client: SMOOR

Jul 2020 – May 2021

- Led implementation of an end-to-end ERP covering procurement, storage, manufacturing, packaging, and order management with real-time tracking.
- Managed and trained a team of freshers; conducted weekly client meetings to keep delivery aligned with business needs.

Tech Stack: Python, MySQL, ERPNext, AWS (EC2, S3, RDS, Lambda, CloudWatch, EventBridge, SNS, SQS)

► **Hostel Management System**

Feb 2020 – Jun 2020

- Sole developer for a microservices-based hostel operations platform — conceptualization through deployment.
- Integrated MongoDB for efficient data management and RabbitMQ for inter-service messaging.

Tech Stack: Flutter, Flask, Microservices, MongoDB, RabbitMQ, AWS (EC2, S3, RDS, Lambda, CloudWatch)

Solution Developer — Next Techno Solutions (P) Ltd

May 2016 – Feb 2020

Coimbatore, Tamil Nadu

► **NIAA Chatbot — AI-Powered HR Assistant**

Mar 2019 – Feb 2020

- Built an HR-enabled conversational chatbot on the Rasa framework, using supervised learning to answer HR and employee queries with accurate, timely responses.
- Collected, curated, and structured the training dataset — designing intents and conversation flows to maximize response accuracy.
- Integrated the bot into Rocket.Chat for seamless employee access across the organization.

Tech Stack: Rasa, Rocket.Chat, Python, JavaScript, MySQL, AWS (EC2, S3, RDS, Lambda)

► **IDAAS — Identity as a Service (SSO Platform)**

Dec 2017 – Mar 2019

- Spearheaded end-to-end implementation of a single sign-on platform providing centralized authentication for all internal web applications.
- Leveraged Keycloak and OpenLDAP to enforce robust security and device-level access control for desktops and laptops.
- Owned the full development lifecycle — from requirement intake to production rollout — meeting all functional and security criteria.

Tech Stack: Keycloak, OpenLDAP, Python

► **EXOS — Cloud Resource Management Platform** · Client: PNG Government

Jul 2016 – Dec 2017

- Implemented core compute, networking, and monitoring functionality for a cloud infrastructure management platform akin to established IaaS consoles.
- Collaborated with the UI team to integrate backend features; optimized backend processes for performance and reliability.
- Integrated multiple modules into the platform with thorough testing and documentation.

Tech Stack: Python, JavaScript, jQuery, Swagger, MySQL, PostgreSQL, OpenNebula, Redis

EDUCATION

Master of Science in Information Technology

2014 – 2016

GAC, Bharathiar University, Coimbatore, Tamil Nadu · CGPA: 7.5